

Research Memo: Neural Network Activation Functions

Emily Huffaker

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Researcher: Emily Huffaker

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Research Question: When using open-ended descriptions to predict dominant transportation mode, which activation function achieves the best F1 score?

Answer, Response, + Summary of Results

To answer this research question, I tested four different activation functions in the neural network's hidden layer while holding the architecture, optimizer (Adam), and training parameters constant. The activation functions tested were ReLU, tanh, Leaky ReLU, and GELU.

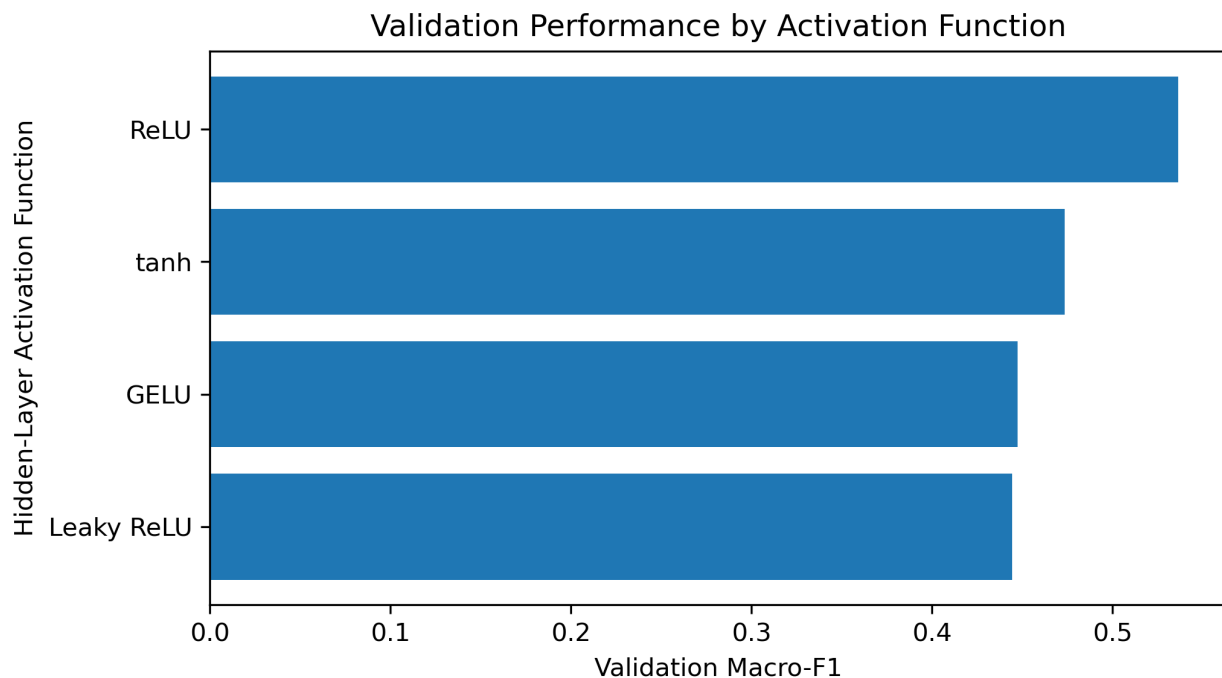


Figure 1: Activation function comparison

The results show that ReLU achieved the highest validation macro-F1 score of 0.536, followed by tanh with 0.474. The complete performance comparison is shown below:

Activation Function	Validation Macro-F1	Validation Accuracy	Epochs Trained
ReLU	0.536	0.667	50
tanh	0.474	0.583	41
GELU	0.447	0.542	47
Leaky ReLU	0.444	0.542	45

These results indicate that ReLU is the most effective choice for this neural network architecture when predicting transportation mode from open-ended text responses.

What Questions or Uncertainties Remain?

Would these activation function rankings remain consistent with different architectures or hyperparameters?

Would the results remain stable across different training, validation, and test splits?

How would these activation functions perform on the apprehension prediction task?